Modeling the Influence Between Music Taste and Internalizing Psychopathology using Bayesian Networks

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Abstract

In this work, we wish to study the relationship between music taste and forms of internalizing psychopathology (internalization). We model Bayesian networks (BN) on a user survey dataset in the domain, using two hypotheses, 'Internalization influences Taste' and 'Taste influences Internalization' and evaluate both with reasoning from the literature and scoring functions applied to the relevant BNs. Our findings imply that modeling influence of internalization over music taste results in a better fitting BN. We shortly explore further BN variants with prior knowledge, latent variables and a learned network.

Introduction

Domain

Internalizing psychopathology, as discussed by (Miranda et al. 2012), relates several different psychopathologies (e.g. Anxiety, Depression, Insomnia, OCD) and opposes them dimensionally to externalizing psychopathology (e.g. aggression) on the basis of correlations, which themselves indicate underlying biopshychosocial factors.

(Miranda et al. 2012) forms a hypothesis based on emotion regulation and gradual development, which implies that music listening might have a small, gradual and cumulative influence on mental health. Possible reasons and mechanisms for this influence are discussed, however, ultimately, this work concludes that empirical studies on the subject are few and difficult to draw conclusions from.

A further analysis of the literature relating to music preference and mental health in adolescents is presented in (Baker and Bor 2008). Interestingly, here the authors conclude from their review that a more likely connection between the two phenomena is that music taste might be influenced by emotional factors.

Aim

Inspired by (Miranda et al. 2012) and (Baker and Bor 2008), we are interested in studying what insight Bayesian networks (BN) can give us into the two hypotheses "Music taste influences internalization" (TiI) and "Internalization influences music taste" (IiT). We use the dataset "Music & Mental Health Survey Results", uploaded to Kaggle by (Rasgaitis). Our main goal is to define and evaluate BNs and perform queries inspired by the hypotheses and the findings of the two papers. Furthermore, we aim to propose different ways of modeling the domain presented by the aforementioned dataset and discuss their advantages.

Method

For each hypothesis we model two basic BNs manually. We evaluate said BNs by means of score functions and perform interesting queries on the more plausible one.

We explore some more ways of modeling the domain by:

1. introducing prior knowledge to account for biases in the aforementioned dataset;
2. adding a latent variable for internalization;
3. applying a score based learning approach to obtain a further network structure.

Results

Following some of the considerations presented in (Baker and Bor 2008) and the results of our quantitative evaluation, we conclude that, at least in some sense, IiT is a more plausible hypothesis and the corresponding BN fits the data better. The queries we perform on this BN give us insight that aligns well with findings in (Miranda et al. 2012).

Our further attempts of modeling the domain show us the advantages of the different approaches and different interpretations of the domain, namely:

1. prior knowledge can account for bias in the dataset;
2. using a latent variable for internalization reduces model complexity;
3. score based structure learning on the dataset suggests variables considered in the hypotheses are independent.

Model

Both of the basic BNs 1 and 2 share a set of edges that are unrelated to the causal direction dilemma and describe listening habits. Otherwise, the causal relations between genres (and features related to genres, such as BPM) and the conditions (Anxiety, Depression, Insomnia, OCD) are mirrored. The CPTs for both were estimated using Maximum Likelihood Estimation (Murphy 2012) on the dataset.

For the model which considers prior knowledge, we rely on data from (Villarroel and Terlizzi 2020), (Terlizzi and

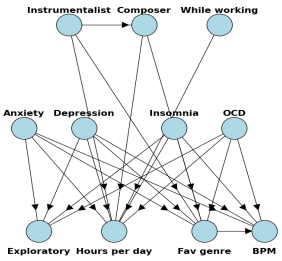


Fig. 1: IiT BN

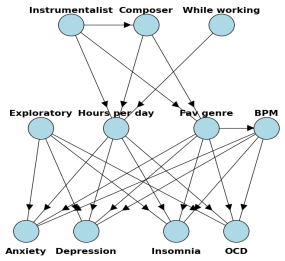


Fig. 2: TiI BN

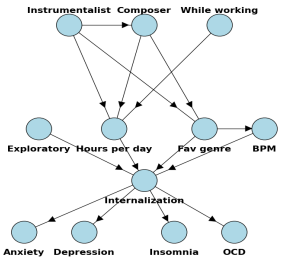


Fig. 3: TiI + Latent Variable

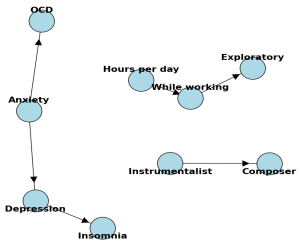


Fig. 4: Learned BN

Villarroel 2020), (Reuben, Elgaddal, and Black 2023) and (Shavitt et al. 2022), which provide information on the prevalence of the internalizing pathologies we consider. In our code we assume for the sake of the example that we've observed 10000 subjects and we obtained distributions similar to those in the above studies. Its structure is the same as the IiT BN, however, in this case we use Bayesian Parameter Estimation which allows us to consider this prior knowledge.

3 is the BN we obtained using a latent variable "Internalization". Its CPTs were estimated using the Expectation Maximization algorithm (Moon 1996), which is designed for incomplete data and is able to handle missing latent variables, on the dataset.

The BN that resulted from score based structure learning 4 was inferred by means of a Hill Climbing search with the BIC score (Koller and Friedman 2009) (Carvalho 2009) functioning as a metric on the dataset.

For information on the variables' exact domains and a discussion on data preprocessing, refer to the *preprocess.ipynb* notebook of the project.

Analysis

Experimental setup

In order to qualitatively compare the two BNs that correspond to our hypothesis dilemma, we use log-likelihood, as well as the two score functions BIC and BDeu (Koller and Friedman 2009) (Carvalho 2009).

	IiT	TiI
BIC	-59170.34	-22026.34
BDeu	-7781.86	-7976.57
Log-Likelihood	-5185.80	-5717.98

Table 1: Comparison of IiT and TiI for the three scoring methods

The results imply that while the BN corresponding to IiT has a higher model complexity it fits the data better. A further consideration in our choice of which BN to use for the performance of queries comes from (Baker and Bor 2008). Said paper draws on conclusions from the existing literature on suicidal ideation, in order to state that music listening habits are likely indicative of existing vulnerabilities, rather than influence them and in its conclusion reiterates that IiT is more likely to be true .This leads us to believe that at least in some sense this hypothesis is more plausible.

We query the IiT BN with a number of questions inspired by (Miranda et al. 2012). The queries are cumulative, and genres are grouped (e.g. Rock, Metal, Hip-Hop) in order to allow for more generalized inference. We evaluate the query results based on comparison with the prior research. The queries, among other things, included favorite genre distributions given the presence of an internalizing condition (or all four), the probability of comorbidity of conditions given a favorite genre, and the correlation between some conditions and habits such as music tempo, hours listened, or listening at work.

Results

Despite the self-reporting nature of the data, we observe some results that align well with prior research (Miranda et al. 2012). Thus, the data reveals comorbidity between anxiety and depression and between insomnia and both. Both conditions are much more prevalent among fans of rock, metal or hip-hop music as opposed to jazz, gospel, or lo-fi music. However, the latter group is significantly more associated with insomnia. There were also some surprising results. First, the reported presence of depression or anxiety does not alter the distribution of the BPM of listened music (even though different genres are associated with different typical tempos, that effect is negligible). Then, the data suggests that people who report depression listen to slightly faster music. In addition, rock was the most common favorite genre in our BN given one of the four conditions, but at a lower rate than in the general data. Finally, when a user reports all four conditions, their most likely favorite genre is metal.

Conclusion

Our study is limited by the insufficient amount of psychological studies on the matter, as well as the subjective nature of the chosen data set. Nevertheless, we are inclined to believe that underlying mental conditions influence music preferences, rather than vice-versa. Our work demonstrated that this relationship exhibits a number of interesting properties that give insight into a number of questions on listening habits, comorbidity, etc. Finally, our short consideration of different ways to model the domain gives insight into the possibilities of further domain exploration using BNs.

Links to external resources

- GitHub repository: https://github.com/kosukach/Music-Taste-and-Mental-Health/
- dataset: https://www.kaggle.com/datasets/catherinerasgaitis/mxmh-survey-results

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